

TASK 1

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import fetch_california_housing
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Load dataset
housing = fetch_california_housing()

# Create dataframe
df = pd.DataFrame(housing.data, columns=housing.feature_names)

# Add target column
df['Price'] = housing.target

# Show first 5 rows
print(df.head())

# Dataset information
print(df.info())
```

```
# Statistical summary
```

```
print(df.describe())
```

```
# Check missing values
```

```
print(df.isnull().sum())
```

```
# Correlation heatmap
```

```
plt.figure(figsize=(10,8))
```

```
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

```
plt.title("Correlation Heatmap")
```

```
plt.show()
```

```
# Features and target
```

```
X = df.drop('Price', axis=1)
```

```
y = df['Price']
```

```
# Train-test split
```

```
X_train, X_test, y_train, y_test = train_test_split(
```

```
    X, y, test_size=0.2, random_state=42
```

```
)
```

```
# Train model
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```

# Predictions
y_pred = model.predict(X_test)

# Evaluation
mae = mean_absolute_error(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
r2 = r2_score(y_test, y_pred)

print("\nModel Evaluation:")
print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)

# Actual vs Predicted plot
plt.figure(figsize=(8,6))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted House Prices")
plt.show()

```

OUTPUT:-

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	Price
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585

```
2 7.2574    52.0 8.288136 1.073446    496.0 2.802260   37.85  -122.24  3.521
3 5.6431    52.0 5.817352 1.073059    558.0 2.547945   37.85  -122.25  3.413
4 3.8462    52.0 6.281853 1.081081    565.0 2.181467   37.85  -122.25  3.422
```

<class 'pandas.DataFrame'>

RangeIndex: 20640 entries, 0 to 20639

Data columns (total 9 columns):

```
# Column    Non-Null Count  Dtype
---  ---
0 MedInc    20640 non-null float64
1 HouseAge  20640 non-null float64
2 AveRooms  20640 non-null float64
3 AveBedrms 20640 non-null float64
4 Population 20640 non-null float64
5 AveOccup  20640 non-null float64
6 Latitude  20640 non-null float64
7 Longitude 20640 non-null float64
8 Price     20640 non-null float64
```

dtypes: float64(9)

memory usage: 1.4 MB

None

	MedInc	HouseAge	AveRooms	...	Latitude	Longitude	Price
count	20640.000000	20640.000000	20640.000000	...	20640.000000	20640.000000	20640.000000
mean	3.870671	28.639486	5.429000	...	35.631861	-119.569704	2.068558
std	1.899822	12.585558	2.474173	...	2.135952	2.003532	1.153956
min	0.499900	1.000000	0.846154	...	32.540000	-124.350000	0.149990

25%	2.563400	18.000000	4.440716	...	33.930000	-121.800000	1.196000
50%	3.534800	29.000000	5.229129	...	34.260000	-118.490000	1.797000
75%	4.743250	37.000000	6.052381	...	37.710000	-118.010000	2.647250
max	15.000100	52.000000	141.909091	...	41.950000	-114.310000	5.000010

[8 rows x 9 columns]

MedInc 0

HouseAge 0

AveRooms 0

AveBedrms 0

Population 0

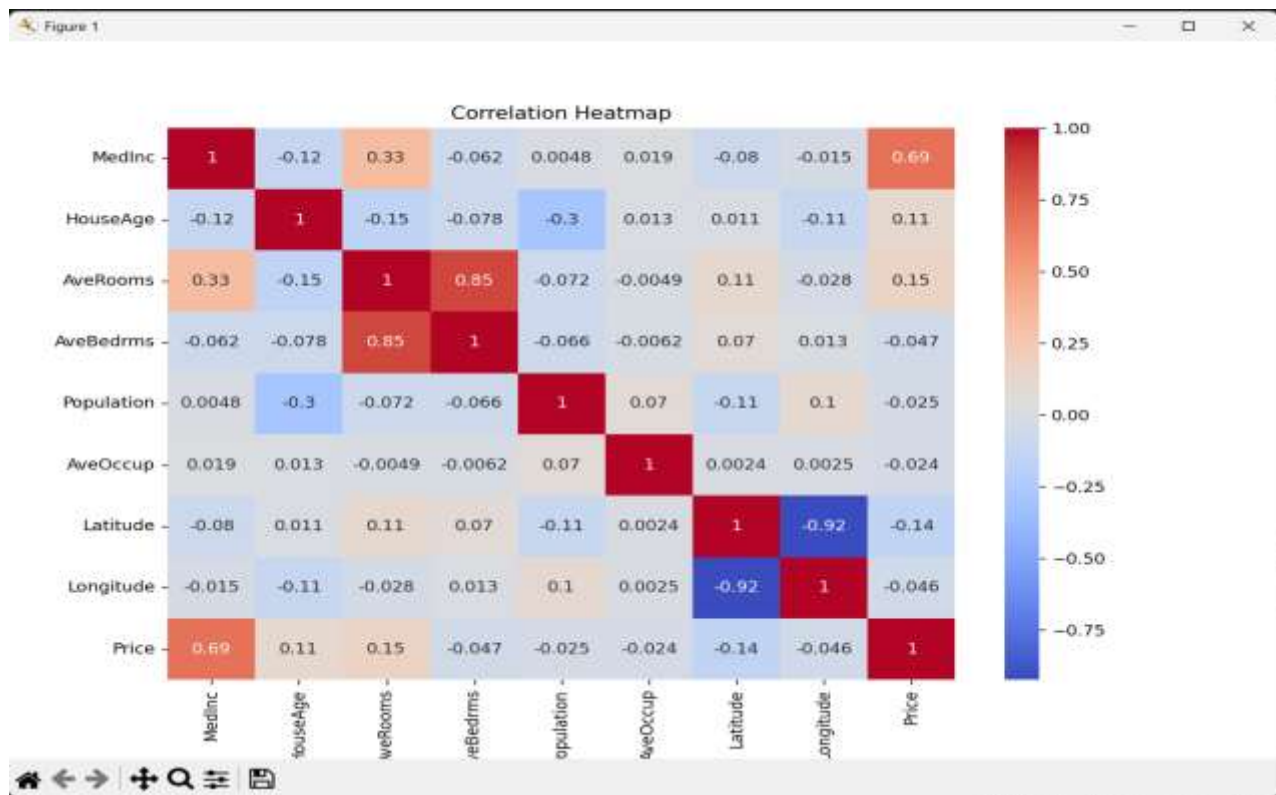
AveOccup 0

Latitude 0

Longitude 0

Price 0

dtype: int64

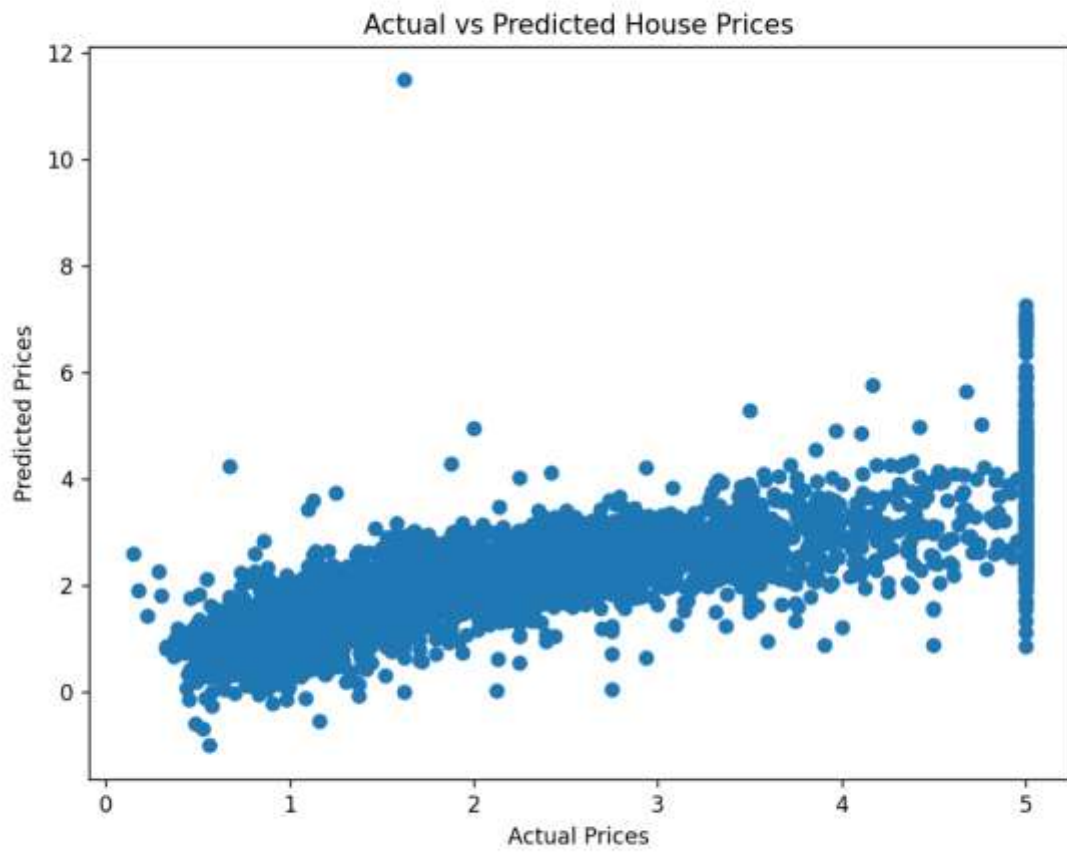


Model Evaluation:

MAE: 0.5332001304956555

RMSE: 0.7455813830127763

R2 Score: 0.575787706032451



-----X-----X-----X-----